Лабораторная работа. Подрисовка изображений

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§ 1. Подрисовка изображений. Постановка задачи.

Под изображением будем понимать вектор $f \in \mathbf{R}^s$. Целью будет нахождение неизвестного изображения $u \in \mathbf{R}^s$ по наблюдаемому изображению f. Пусть $\Omega = \{1, 2, \ldots, s\}$ и $\Lambda \subset \Omega$. Наблюдаемое изображение имеет вид:

$$f_i = \begin{cases} u_i + \eta_i, & i \in \Lambda; \\ \text{проивольно}, & i \in \Omega \setminus \Lambda. \end{cases}$$

Задача восстановления неизвестного изображения u относится к задаче аппроксимации. Обозначим P_Λ диагональную матрицу, у которой $(\mathrm{P}_\Lambda)_{i,i} = 1, \ \ \forall i \in \Lambda$ и $(\mathrm{P}_\Lambda)_{i,i} = 0, \ \ \forall i \in \Omega \setminus \Lambda$. Кроме того, введем еще следующие обозначения: $W: \ f \to Wf$ -- дискретное вейвлет-преобразование (вейвлет-восстановление), ставящее в соответствие данному вектору вейвлет-коэффициентов $\beta \in \mathbf{R}^d$ изображение $f = W\beta \in \mathbf{R}^s$ и W^{-1} -- обратное преобразование. Пусть

$$\Gamma_\lambda(\beta_1, \ldots, \beta_d) = (\, t_{\lambda_1}(\beta_1) \quad \ldots \quad t_{\lambda_d}(\beta_d)\,)\,,$$

где

$$t_{\lambda_i}(\beta_i) = \begin{cases} \mathrm{sign}(\beta_i)(|\beta_i| - \lambda_i), & |\beta_i| > \lambda_i, \\ 0, & |\beta_i| \leqslant \lambda_i. \end{cases}$$

Изображение u будем находить из следующей задачи минимизации:

$$u = W\beta^*;$$

$$\beta^* = \min_{\beta \in \mathbf{R}^d} \frac{1}{2} \|\mathbf{P}_\Lambda W\beta - \mathbf{P}_\Lambda f\|_2^2 + \mu \|\beta\|_1.$$

Алгоритм решения оптимизационной задачи

1. Выберем произвольный вектор $\beta_0 \in \mathbf{R}^d$

2. Находим $\beta_{k+1} = \Gamma_\mu \left(W\mathrm{P}_\Lambda f + \beta_k - W\mathrm{P}_\Lambda W^{-1}\beta_k\right)$, $k = 0, 1, \ldots$

Литература

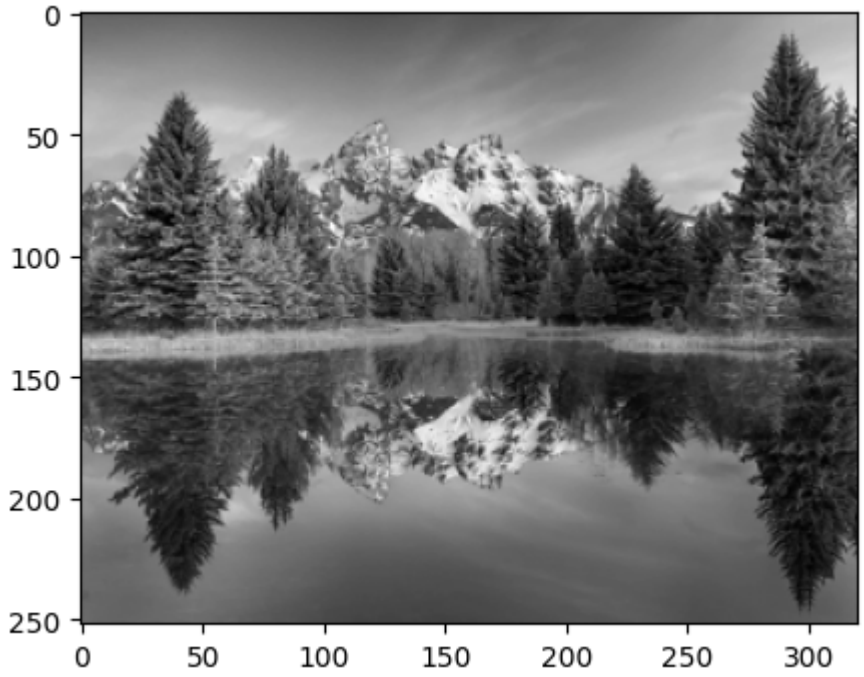
1. J.F. Cai, R.H. Chan, and Z. Shen,A framelet-based image inpainting algorithm, Applied and Computational Harmonic Analysis24(2008), no. 2, 131–149.

In [31]:
```
import numpy as np
from PIL import Image
import matplotlib.pyplot as plt
import pywt
%matplotlib inline
```

In [55]:
```
f = Image.open('im_7.jpg').convert('L')
f = f.resize((320, 252))
print(f.size)
fig = plt.figure(figsize = (5,5))
plt.imshow(f, cmap='gray')
```

```
(320, 252)
```

Out[55]:
```
<matplotlib.image.AxesImage at 0x7b90822c06e0>
```



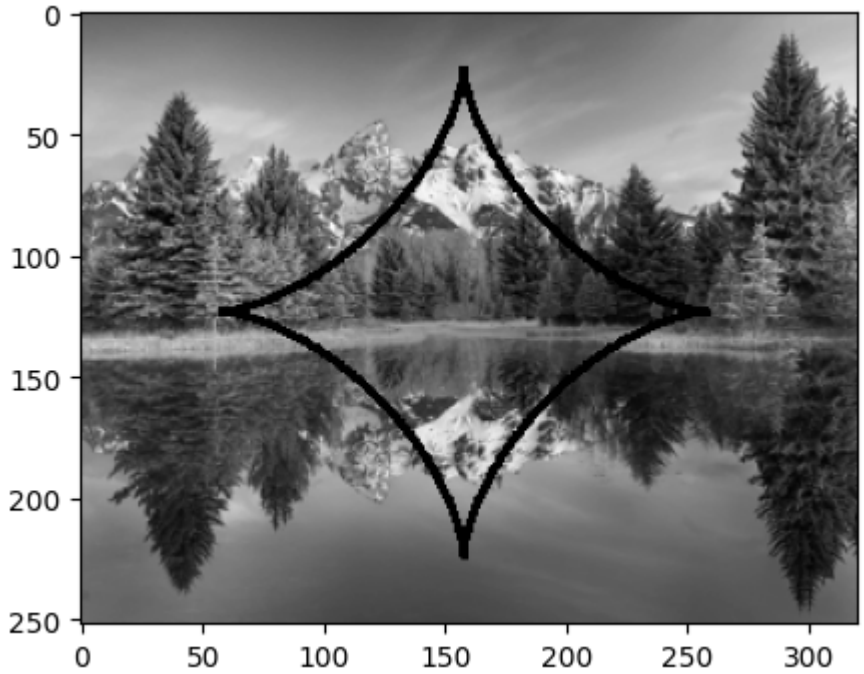
In [56]:
```
coeffs = pywt.wavedecn(f, wavelet = 'db4', level = 2)
arr, coeff_slices = pywt.coeffs_to_array(coeffs)
coeffs_from_arr = pywt.array_to_coeffs(arr, coeff_slices, output_format = 'wavedecn')
I = pywt.waverecn(coeffs_from_arr, wavelet = 'db4')
print(np.max(np.abs(f - I)))
```

```
2.2737367544323206e-13
```

In [57]:
```
''' Здесь создаем то, что будем закрашивать. Удалять будем линии '''
P_Lambda = np.zeros((np.array(f).shape)) + 1
x0 = np.int64(np.array(f).shape[0] / 2)
y0 = np.int64(np.array(f).shape[1] / 2)
N = 1000
for i in range(N):
    x = np.int64(x0 + 100 * ((np.cos(2 * np.pi * i / N)) ** 3))
    y = np.int64(y0 + 100 * ((np.sin(2 * np.pi * i / N)) ** 3))
    P_Lambda[x-3:x, y-3:y] = 0
```

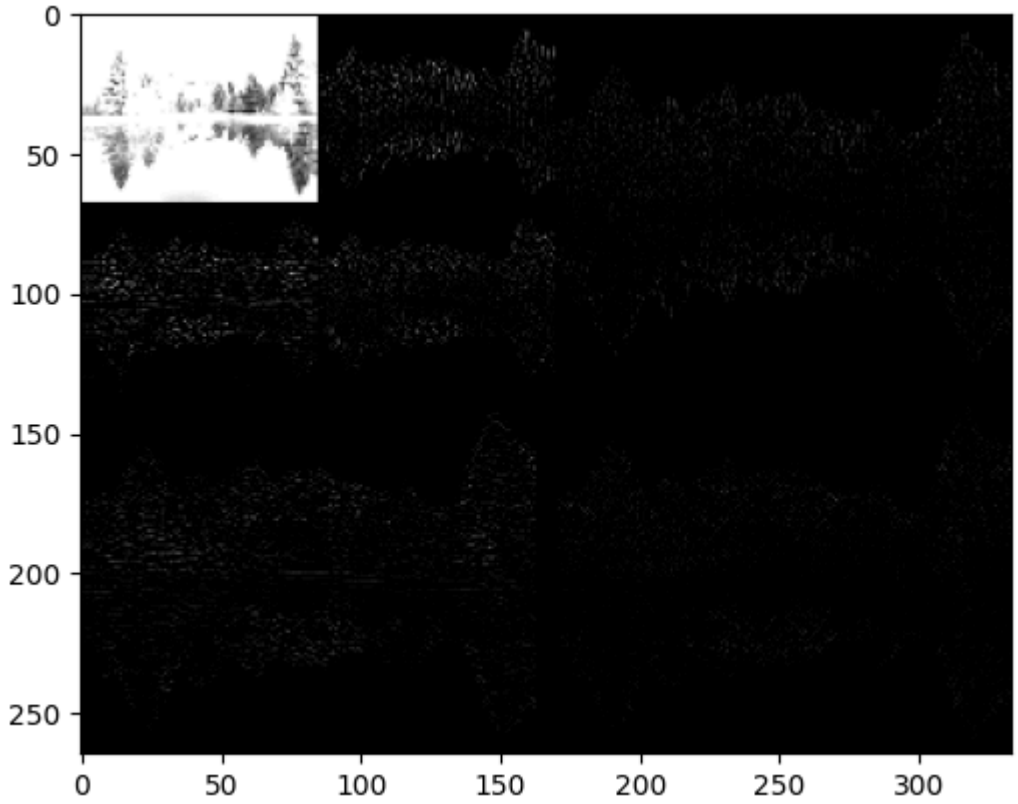
In [58]:
```
Pf = P_Lambda * np.array(f)
fig = plt.figure(figsize = (5,5))
plt.imshow(Pf, cmap='gray')
print(Pf.shape)
```

```
(252, 320)
```



In [59]:
```
plt.imshow(Image.fromarray(arr))
```

Out[59]:
```
<matplotlib.image.AxesImage at 0x7b9082410530>
```



Алгоритм решения оптимизационной задачи

$$\beta^* = \min_{\beta \in \mathbf{R}^d} \frac{1}{2} \|\mathbf{P}_\Lambda W \beta - \mathbf{P}_\Lambda f\|_2^2 + \mu \|\beta\|_1.$$

1. Выберем произвольный вектор $\beta_0 \in \mathbf{R}^d$
2. Находим $\beta_{k+1} = \Gamma_\mu \left(W \mathrm{P}_\Lambda f + \beta_k - W \mathrm{P}_\Lambda W^{-1} \beta_k \right)$, $k = 0, 1, \ldots$

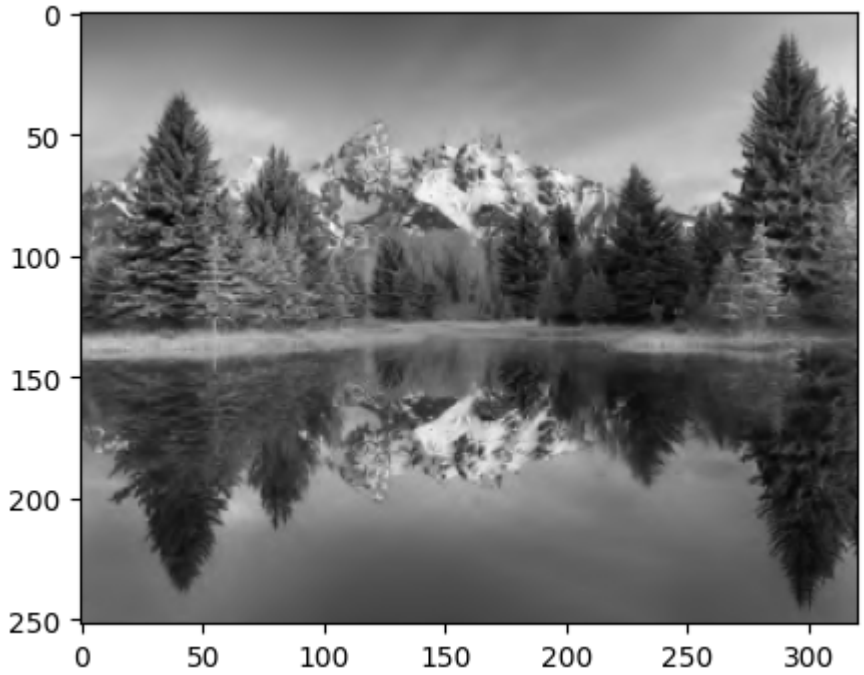
In [60]:
```
def Gamma(A, lamb):
    B = np.abs(A) - lamb
    B[np.nonzero(B <= 0)] = 0
    B = np.sign(A) * B
```

```
    return B


def Bregman_Lin(Pf, P_Lambda, Epoch, mu, Level, wavelet):
    beta_k = pywt.wavedecn(Pf, wavelet=wavelet, level=Level)
    arr_beta_k, coeff_slices_beta_k = pywt.coeffs_to_array(beta_k)
    beta_k = arr_beta_k.flatten()

    coeffs_P_Lambda = pywt.wavedecn(P_Lambda, wavelet=wavelet, level=Level)
    arr_P_Lambda, coeff_slices_P_Lambda = pywt.coeffs_to_array(coeffs_P_Lambda)
    P_Lambda_wavelet = arr_P_Lambda.flatten()

    coeffs_Pf = pywt.wavedecn(Pf, wavelet=wavelet, level=Level)
    arr_Pf, coeff_slices_Pf = pywt.coeffs_to_array(coeffs_Pf)
    Pf_wavelet = arr_Pf.flatten()

    def W(beta, coeff_slices, output_format):
        arr_beta = beta.reshape(arr_beta_k.shape)
        coeffs = pywt.array_to_coeffs(arr_beta, coeff_slices, output_format=output_format)
        return pywt.waverecn(coeffs, wavelet=wavelet)

    def W_inv(f):
        coeffs = pywt.wavedecn(f, wavelet=wavelet, level=Level)
        arr, coeff_slices = pywt.coeffs_to_array(coeffs)
        return arr.flatten(), coeff_slices, 'wavedecn'

    for k in range(Epoch):
        Wbeta_k = W(beta_k, coeff_slices_beta_k, 'wavedecn')
        P_Lambda_Wbeta_k = P_Lambda * Wbeta_k
        W_inv_P_Lambda_Wbeta_k, _, _ = W_inv(P_Lambda_Wbeta_k)

        W_P_Lambda_f, _, _ = W_inv(P_Lambda * Pf)

        beta_k_reshaped = beta_k.reshape(arr_beta_k.shape)
        arg = W_P_Lambda_f + beta_k - W_inv_P_Lambda_Wbeta_k
        beta_k = Gamma(arg, mu)

    u = W(beta_k, coeff_slices_beta_k, 'wavedecn')
    return u
```

In [61]:
```
Epoch = 100
mu = 6
delta = 0.01
Level = 4
wavelet = 'db4'
u = Bregman_Lin(Pf, P_Lambda, Epoch, mu, Level, wavelet)
```

In [62]:
```
fig = plt.figure(figsize = (5,5))
plt.imshow(u, cmap='gray')
```

Out[62]:
```
<matplotlib.image.AxesImage at 0x7b9081cf8380>
```

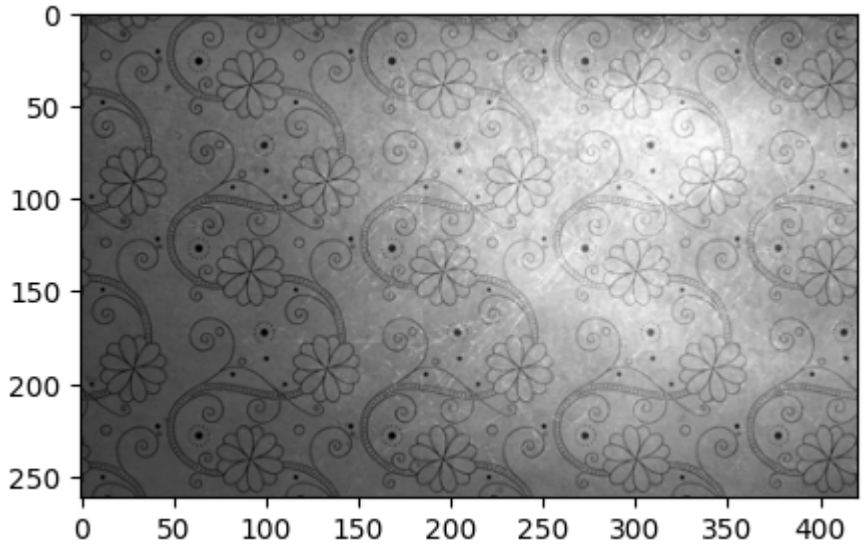


§ 2. Удаление текста

In [63]:
```
f = Image.open('im6.jpeg').convert('L')
h, w = f.size
f = f.resize((np.int64(h / 4), np.int64(w / 4)))
```

```
h, w = f.size
fig = plt.figure(figsize = (5,5))
plt.imshow(f, cmap='gray')
print(f.size)
```

```
(420, 262)
```

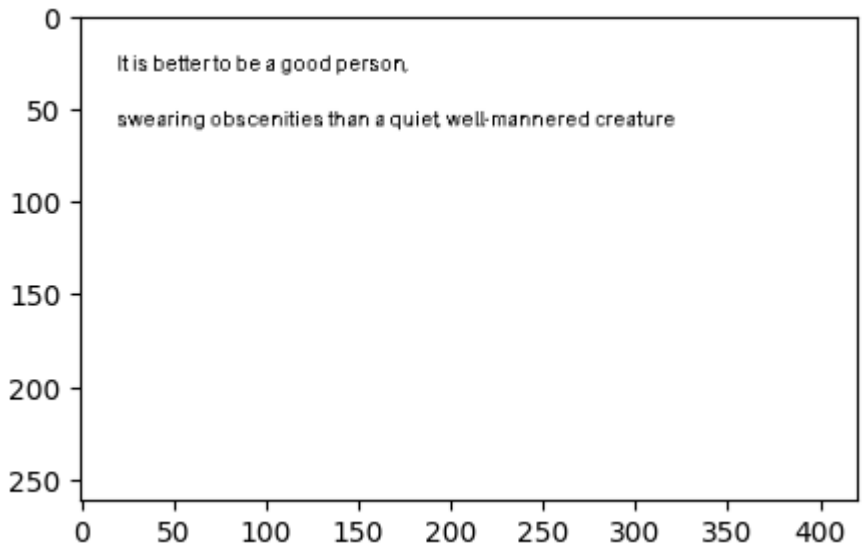


In [64]:

```
from PIL import ImageDraw
f_text = Image.fromarray(np.zeros((w, h)) + 255)
draw = ImageDraw.Draw(f_text)
draw.text((20, 20), "It is better to be a good person,")
draw.text((20, 50), "swearing obscenities than a quiet, well-mannered creature")
fig = plt.figure(figsize = (5,5))
plt.imshow(f_text, cmap='gray')
```

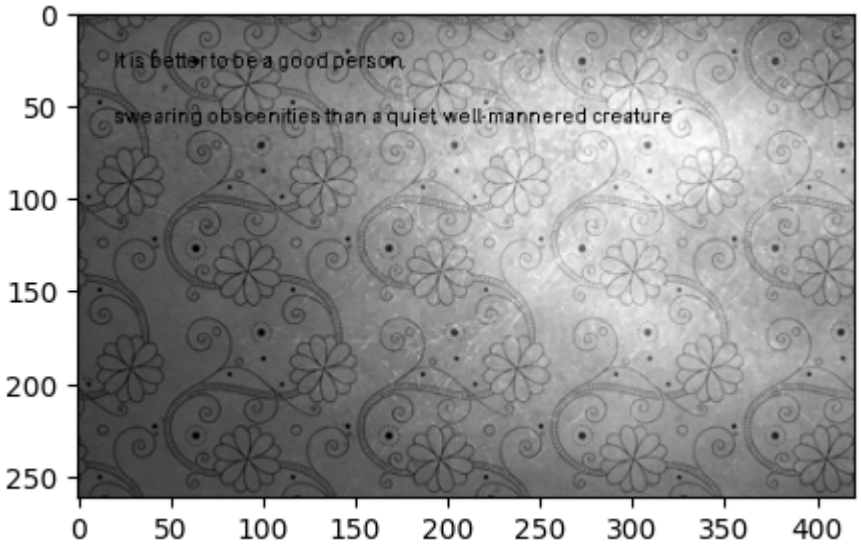
Out[64]:

```
<matplotlib.image.AxesImage at 0x7b9081babd10>
```



In [65]:
```
f = np.array(f)
P_Lambda = np.zeros((f.shape)) + 1
T = np.nonzero(np.array(f_text) < 255)
P_Lambda[T] = 0
Pf = P_Lambda * f
fig = plt.figure(figsize = (5,5))
plt.imshow(Pf, cmap='gray')
```

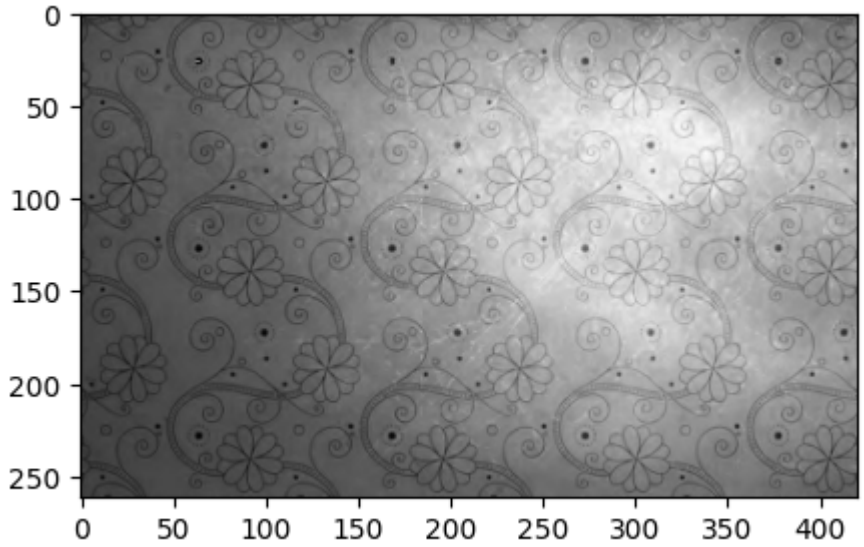
Out[65]:
```
<matplotlib.image.AxesImage at 0x7b9081b81e50>
```



In [66]:
```
Epoch = 100
mu = 6
delta = 0.01
Level = 4
wavelet = 'db4'
u = Bregman_Lin(Pf, P_Lambda, Epoch, mu, Level, wavelet)
```

In [67]:
```
fig = plt.figure(figsize = (5,5))
plt.imshow(u, cmap='gray')
```

Out[67]:
```
<matplotlib.image.AxesImage at 0x7b9081a6fbf0>
```


0
50
100
150
200
250
0
50
100
150
200
250
300
350
400